

Cloud computing

Understanding Cloud Storage Mechanisms

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What is Cloud Storage?

- Stores data on remote servers over the Internet
- Managed by cloud service providers
- Accessible from anywhere with an Internet connection
- Supports collaboration and file sharing
- Reduces dependence on local storage

How Data is Stored

- Data is divided into blocks or objects
- Stored across multiple servers and data centres
- Replication creates multiple copies for reliability
- Compression reduces storage space
- Redundancy prevents data loss

How Data is Managed

- Metadata organises files efficiently
- Automatic backup and recovery
- Encryption protects sensitive data
- Version control maintains previous file copies
- Load balancing improves performance

How Data is Accessed

- Access using web browsers, mobile apps, or APIs
- Authentication verifies user identity
- Role-based permissions control access
- Fast retrieval using distributed servers
- Real-time synchronisation across devices

Types of Cloud Storage

- Object Storage – Images, videos, backups
- File Storage – Shared documents and folders
- Block Storage – Databases and virtual machines

Advantages

- Scalable storage capacity
- Cost-effective
- High availability
- Disaster recovery
- Global accessibility
- Secure sharing

Challenges

- Requires Internet connection
- Privacy and security concerns
- Vendor lock-in
- Possible service outages
- Compliance issues

Popular Cloud Storage Services

- Google Drive
- Microsoft OneDrive
- Dropbox
- Amazon S3
- Apple iCloud
- Box

Conclusion

- Cloud storage provides secure, scalable, and reliable storage.
- It enables easy access, backup, and collaboration.
- It is a key technology in modern cloud computing.